Ch 2.1: What is Statistical Learning?

Lecture 2 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Weds, Jan 14 , 2026

Announcements

Last time:

- Discussed where to find everything
  - Course webpage
  - D2L
- Check out the syllabus!

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression		HW #2 Due Sun 2/1		
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			

Announcements:

- First homework due Sun Jan 25
- First office hours next week

Covered in this class

- Input/output variables
- Prediction vs inference
- Reduceable vs irreduceable error
- Overfitting
- Classification vs regression
- Supervised vs Unsupervised learning

- Please note: no jupyter notebook for today's class, slides only

An example data set: `Advertising`

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1	4.8
10	199.8	2.6	21.2	10.6
11	66.1	5.8	24.2	8.6

- Sales of a product in 200 markets, along with amount spent on three differnt types of advertising
- Goal: Predict Sales based on amount spent in each type of advertising
- Input variables: TV, Radio, Newspaper
- Output variable: Sales

Data available at msu-cmse-courses.github.io/CMSE381-S26/DataSets/DataSets.html

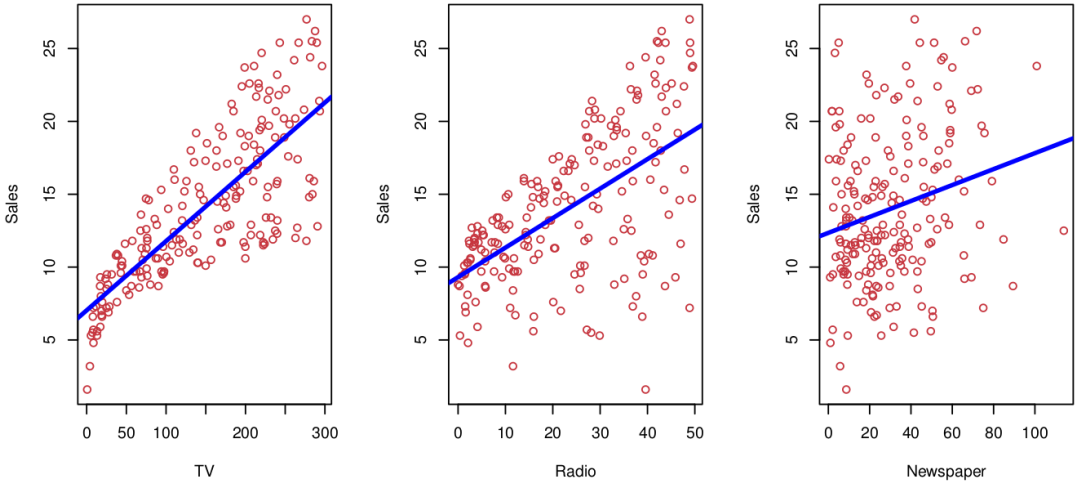
Notation and Big Assumption

Input variables: $X_1, X_2, \cdots, X_p$
Output variable: Y

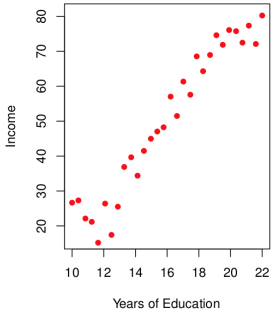
$$Y = f(X) + \varepsilon$$

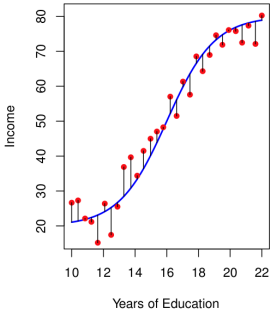
- f is the systemic information that X provides about Y.
- It is the ground truth that we want but can't access
- So our goal is to come up with an estimated model $\hat{f}$.
- ε is a random error term which is independent of X and has mean 0

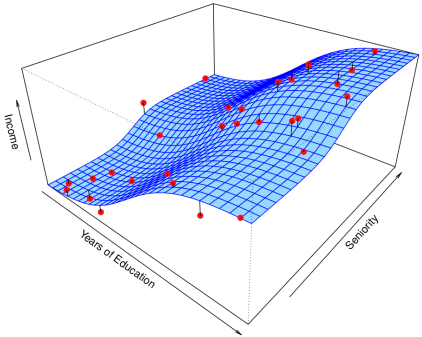
Advertising Example



More examples







Section 1

Prediction vs Inference

Prediction

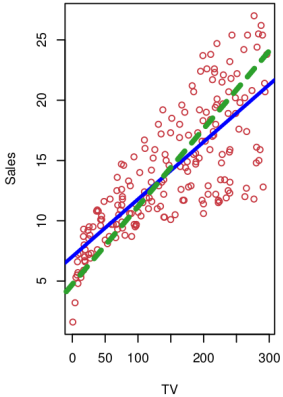
Given a value X, try to provide an estimate for $f(X)$.

Build a model:

$$\hat{Y} = \hat{f}(X)$$

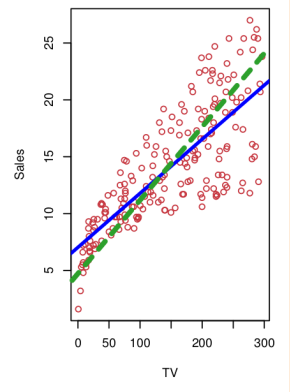
- Want to get a good guess for f, which is unknown blue
- Model is $\hat{f}$ is green dashed lines

Example: If we spend $250 on TV advertising, what do we predict we will we make in sales?



Group question:

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3



The blue solid line is f. The green dashed line is $\hat{f}$.

- What is the predicted sales for the first three data points using the green dashed line $\hat{f}$ shown in the graph?
  - Note all values approximate
  - $\hat{f}(230.1) = 19$,
  - $\hat{f}(44.5) = 7$,
  - $\hat{f}(17.2) = 5$,
- Using the dashed green line as the predicted model $\hat{f}$, what is the error in each of the three predicions?

Reduceable vs irreducable error

All models are wrong, some are useful.

$$Y - \hat{Y}$$

Reducible Error

- $\hat{f}$ will not be a perfect estimate for f.
- We can potentially improve the irreducible accuracy of $\hat{f}$ by using the most appropriate statistical learning technique

Irreducible Error

- Model was $Y = f(X) + \varepsilon$,
- Variability of ε also affects predictions
- Not matter how well we estimate f, we can't get rid of this error.
- Would expect this in real life though:

More on error

- Given estimate $\hat{f}$ (fixed)
- Set of predictors X (fixed)
- Prediction $\hat{Y} = \hat{f}(X)$

$$E(Y - \hat{Y})^2 = E[f(X) + \varepsilon - \hat{f}(X)]^2 = [f(X) - \hat{f}(X)]^2 + Var(\varepsilon)$$

$[f(X) - \hat{f}(X)]^2$ is reducible, $Var(\varepsilon)$ is irreducible

Inference

Want f, but not for prediction (or possibly combined with prediction)

- Which predictors are associated with the response?
- What is the relationship between the response and each predictor?
- Can the relationship between Y and each predictor be adequately summarized using a linear equation? Is it more complicated?

Determine whether each scenario is prediction, inference, or both.

Application	Prediction	Inference
Predict effectiveness of vaccine		
Determine the address written on the image of an envelope.		
Identify risk factors for getting long covid.		
Transcribe an audio file of a person talking.		
Predict stock prices.		

Section 2

How to estimate f?

Input: Training data

- n data points observed
- x_{ij} is the jth predictor for observation i
- y_i is the response variable for the ith observation
- Training data:
  - $\{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$
  - $x_i = (x_i, x_{i2}, \cdots, x_{ip})^T$

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1	4.8
10	199.8	2.6	21.2	10.6
11	66.1	5.8	24.2	8.6

Parametric methods

Step 1: Select a model

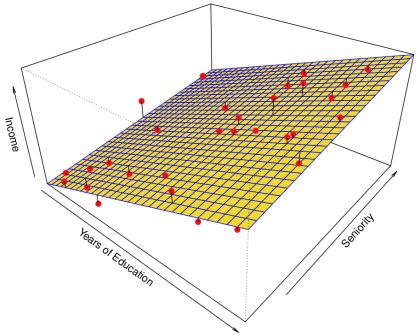
Example:

$$f(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

Step 2: Train the model

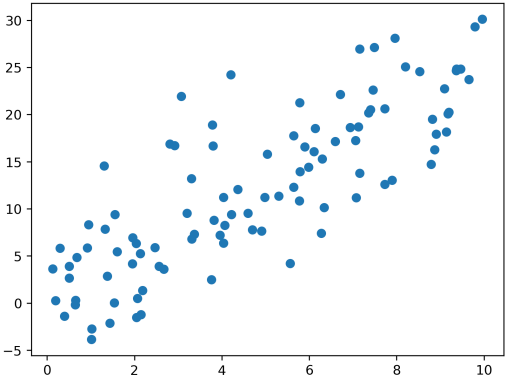
Example:
Find $\beta_i's$ so that

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$



How do you decide on the coefficients?

$$Y \approx \beta_0 + \beta_1 X_1$$

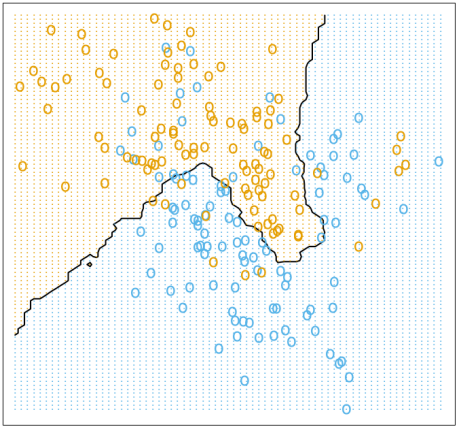


Desmos toy: https://www.desmos.com/calculator/skvt8c73l7

Example Non-parametric method: Nearest Neighbors

$$N_k(x) = \text{Set of } k \text{ nearest neighbors of } x$$

$$\hat{f}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$



$k = 15$

Parametric methods: Pros and Cons

Pros

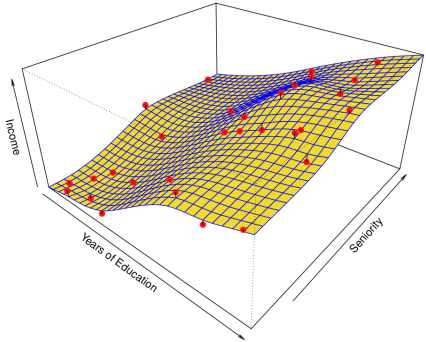
- Easier to estimate paramters than to figure out a completely arbitrary function

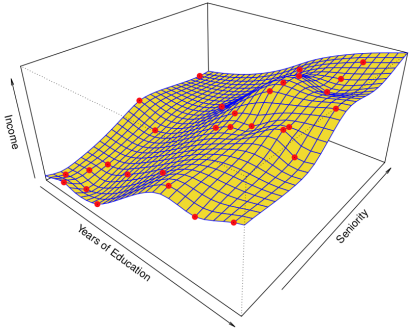
Cons

- You might have chosen the wrong function type

Overfitting

Possible fix: Find more **Flexible** models, which means braider functional form
Problem: needs more variables, could lead to overfitting
Overfitting: Following the noise too closely

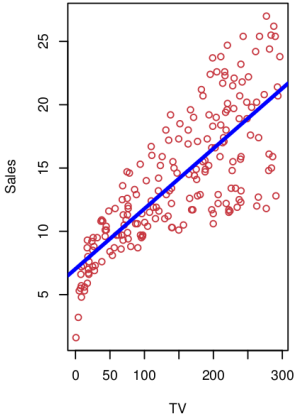




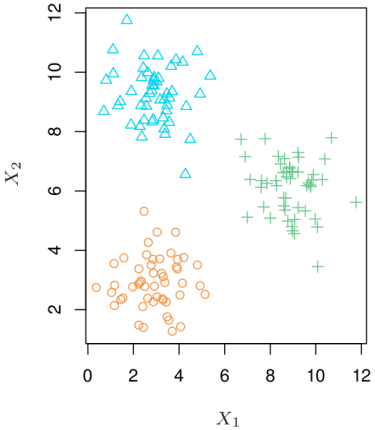
Prediction Accuracy vs Model Interpretability

- More flexible allows for greater accuracy, but potential for overfitting
- Also more restrictive makes it easier to understand and interpret the results

Supervised learning:
Training data has response variable y for every input x



Unsupervised Learning:
Training data does not have response variable y for every input x



Regression vs Classification

Types of variables: Emphasize this is output variable, and which it is determines regression vs classification

- Quantitative
  Ex: Blood pressure, temperature, volume, height, income

- Qualitative / Categorical
  Purchased a ticket, owns a house, Job, Digit in MNIST,

Section 3

Group work

(a) We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

From Ex 2.4.2

- Is this classification or regression? regression
- Do we want inference or prediction? Inference
- What is n, the number of data points? 500
- What is p, the number of variables? 3

(b) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

- Is this classification or regression? classification
- Do we want inference or prediction? Prediction
- What is n, the number of data points? 20
- What is p, the number of variables? 13

TL;DR

- Input/output variables
- Prediction vs inference
- Reduceable vs irreduceable error
- Overfitting
- Classification vs regression
- Supervised vs Unsupervised learning

Wrap up

Next time:

- Friday 1/16
  - Bring Laptop!
  - First homework due Sun Jan 25
  - There will be a quiz this Friday

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression		HW #2 Due Sun 2/1		
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			
10	W	2/4	Logistic Regression	4.1, 4.2, 4.3.1-3		Q4	

Announcements:

- Office hours!
  - Dr. Bao: Tue 9 am-11 am (EB 2507 L)
  - Dr. Khasawneh: MWF 13:40 pm-2:00 pm (EB 2400), Tue 1:30 pm-2:30 pm (Zoom)
  - Siyu : Wed 2:00 pm-3:00 pm (EB 2504)
  - Haishen: WF 2:00 pm-3:00 pm (EB2504)